

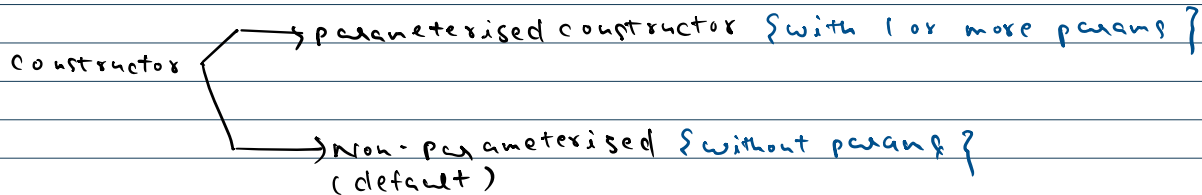
Constructor

It is used to initialise attributes of the class.

Static
block

Constructor

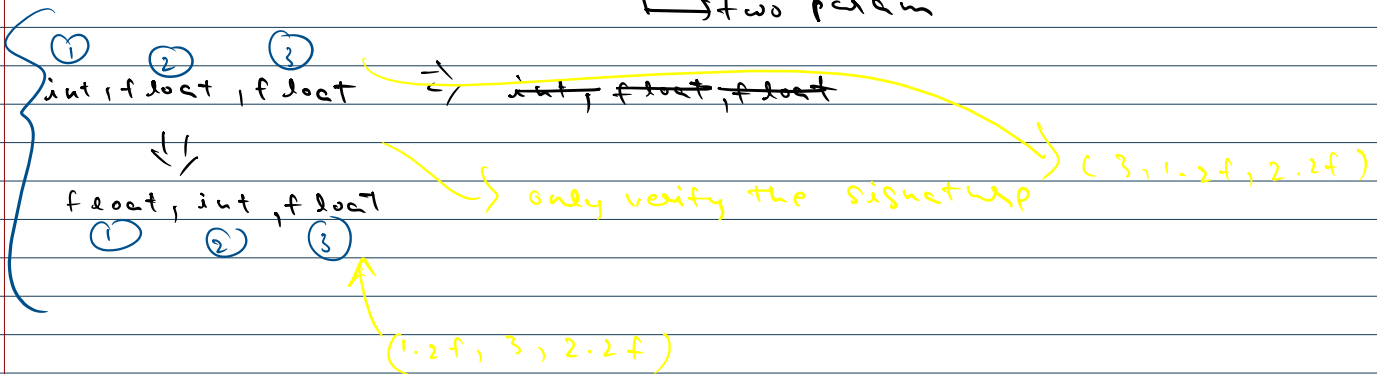
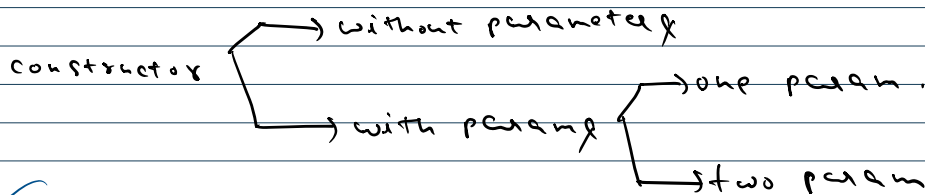
any object if we form,
it will be called



Multiple constructors can be defined in the same class.

Polymorphism

ability to take multiple forms



Step by Step Java Execution

variables with
default values

parsed/initialised
values will
be assigned

constructor
called

```

4 usages
public class Person2 {
3 usages
  String name = "Random";
4 usages
  int age = 91;
1 usage
}

1 public class Person2Client {
2   public static void main(String args[]) {
3     Person2 p = new Person2();
4     p.age = 25;
5     p.name = "Akarsh";
6
7     p.age = 26;
8
  }
}
  
```

JVM

7K

9K

9K?

name → Random

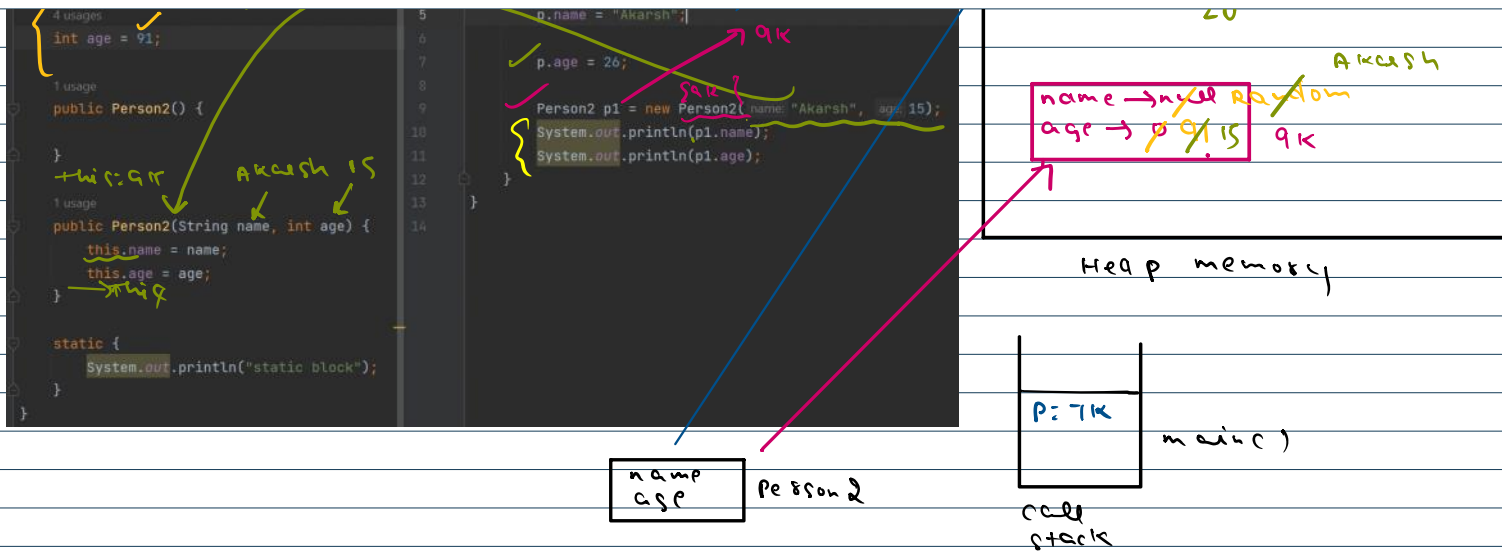
age → 91

26

Akarsh

Akarsh

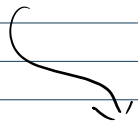
name → Random



class Naming convention.

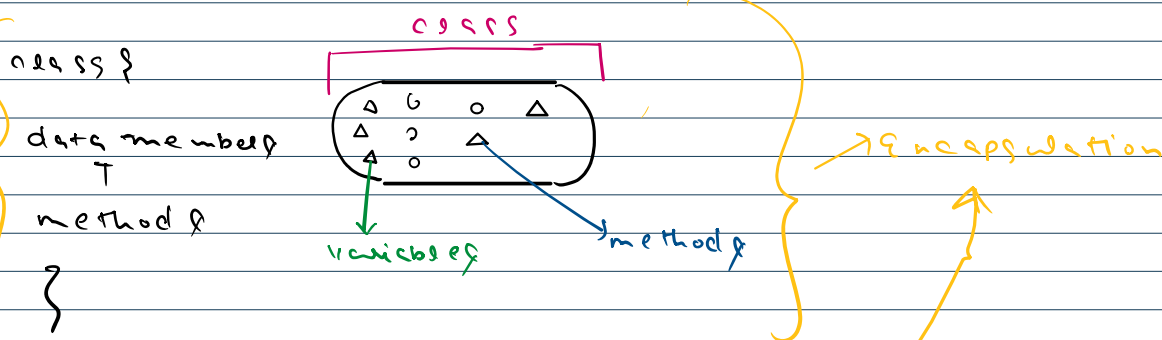
Encapsulation

It is wrapping of the data and code.



restrict the direct access of some of the object components.

1. Protects data
2. Data validation
3. Easier to use/maintain/extend



Access modifiers in Java

public accessible for all classes

private only accessible in declared class

public accessible for all classes
 private only accessible in declared class
 default in same class
 (no modifier) +
 all classes in that package
 protected within the same package
 +
 subclasses (even if they are not in
 the same package)

Exception Handling

Exception

It is an unwanted or unexpected event, which mainly occurs at runtime.

It disrupts the normal flow of program.

Handling $\Rightarrow$ ensuring that normal flow is not interrupted.

Error vs Exception

Error (blue arrow) $\rightarrow$ out of memory, stack overflow
 Exception (green arrow) $\rightarrow$ null pointer exception, IO Exception

